

HW6

- **Submission date: 04/14/2025 (11:59 PM)**
 - Read your class notes *thoroughly* before attempting the assignment.
 - You can submit **well commented** code in either `.py` or `.ipynb` formats.
 - Submitting a PDF explaining your code is *highly* encouraged.
1. **Logic Gates.** In class, we discussed that the XOR gate could be realized with a multi-layer feedforward network by observing that for any two binary inputs X_1 and X_2 , $X_1 \text{ XOR } X_2 = (X_1 \text{ AND } (\text{NOT } X_2)) \text{ OR } ((\text{NOT } X_1) \text{ AND } X_2)$. Draw the complete network for the XOR gate showing all weights and biases for each neuron in the network.
 2. **Three-input gates.** Suppose I wish to implement $X_1 \text{ AND } X_2 \text{ AND } X_3$. Can I realize this gate using (a) two neurons each with 2 inputs and a bias, and (b) a single neuron with 3 inputs and a bias? Whenever possible, draw the network with all weights and activation functions.
 3. **Bagging.** In this exercise, we will numerically estimate the probability that a given observation is part of a bootstrap sample (**reference:** Sec. 5.2 and 5.4, ISLP).
 - a. Generate a set $\mathcal{D}$ of $n = 1,000$ random Gaussian observations and tag observation $j = 4$. Generate 200 bootstrapped samples $\mathcal{D}^{(i)}$, $i = 1, 2, \dots, 200$, from this set and find the fraction of bootstrapped samples that do not contain the tagged observation.
 - b. Using the above procedure, create a plot that displays, for each integer value n from 100 to 100,000, the fraction of bootstrapped samples that contain the tagged (j^{th}) observation.
 - c. Given that $\lim_{n \rightarrow \infty} \left(1 - \frac{1}{n}\right)^n = \frac{1}{e}$, comment on what your plot in Part b. above looks like in for large values of n .
 4. **Boosting.** Complete the Scikit-Learn exercise on using the AdaBoost Classifier.